Chapter 6

Functions

6.1 Basic Definitions

Definition: Let X, Y be two sets. A ***function*** f from X to Y, denoted by $f : X \to Y$, is a relation from X to Y with the following properties:

(i) $Dom(f) = X$;
(ii) $(x, y) \in f \wedge (x, z) \in f \Rightarrow y = z$.

Property (i) implies that for every $x \in X$, there is a $y \in Y$ such that $(x, y) \in f$ and Property (ii) implies that such a y is unique. The unique y is customary denoted by $f(x)$, and $(x, y) \in f$ is customary written as $y = f(x)$. Property (ii) can thus be restated as $(f(x) = y \wedge f(x) = z \Rightarrow y = z)$.

Definition: Let $f : X \to Y$ and $(x, y) \in f$, or equivalently, $y = f(x)$. Then f is said to ***map x onto y***, or y is the ***image*** of x under f.

Example 6.1.18

(i) Let $X = \{a, b, c, d\}, Y = \{x, y, z\}$ and $Z = \{\alpha, \beta, \gamma, \delta\}$.

Let $f_1 = \{(a, x), (b, y), (c, z)\}$, $f_2 = \{(a, x), (b, y), (c, x), (d, x)\}$, $f_3 = \{(a, x), (b, y), (c, z), (d, z)\}$ and $f_4 = \{(a, \alpha), (b, \beta), (c, \gamma), (d, \delta)\}$.

The relation f_1 is not a function from X to Y because $Dom(f) \neq X$. However, it is a function from the set $\{a, b, c\}$ to Y.

The relations f_2 and f_3 are functions from X to Y.

The relation f_4 is a function from X to Z.

(ii) The relation $\emptyset$ is a function from $\emptyset$ to any set Y.

Proof: (i) $Dom(\emptyset) = \emptyset$. (Example 5.1.2(iv))

(ii) (Vacuous Proof))

$$(x, y) \in \emptyset \wedge (x, z) \in \emptyset \Rightarrow false \wedge false \quad \text{(Lemma 3.5.7)}$$
$$\Rightarrow false \quad \text{(E3)}$$

Hence, $(x, y) \in \emptyset \wedge (x, z) \in \emptyset \Rightarrow y = z$

From (i) and (ii) above, we conculde that $\emptyset$ is a function from $\emptyset$ to Y. □

(iii) The congruence relation $\equiv_m$ is not a function from $\mathbf{Z}$ to $\mathbf{Z}$. This is because $0 - m = (-1)m$ and $0 - 2m = (-2)m$, and $-1, -2 \in \mathbf{Z}$ imply that $(0, m), (0, 2m) \in \equiv_m$. But $m \neq 2m$. So, Property (ii) does not hold for $\equiv_m$. As a result, $\equiv_m$ is not a function.

(iv) Let $\mathbf{N}$ be the set of all positive integers and $\mathbf{E}$ be the set of all positive even integers.
The relation $f = \{(x, y) \in \mathbf{N} \times \mathbf{E} \mid y = 2x\}$ is a function from $\mathbf{N}$ to $\mathbf{E}$.

Proof: (i) Since f is a relation from $\mathbf{N}$ to $\mathbf{E}$,

$Dom(f) \subseteq \mathbf{N}$ (Definition of domain of relation). . . . (I)

It remain to provet that $\mathbf{N} \subseteq Dom(f)$.

Let $x \in \mathbf{N}$. Then $2x \in \mathbf{E}$ (Definition of $\mathbf{E}$)

$\Rightarrow (x \in \mathbf{N}) \wedge (2x \in \mathbf{E})$ (I6)

$\Rightarrow (x, 2x) \in \mathbf{N} \times \mathbf{E}$ (Definition of $\times$)

$\Rightarrow (x, 2x) \in f$ (Definition of f)

$\Rightarrow x \in Dom(f)$ (Definition of domain)

We thus have: $x \in \mathbf{N} \Rightarrow x \in Dom(f)$

$\Rightarrow (\forall x)(x \in \mathbf{N} \Rightarrow x \in Dom(f))$ (Gen)

$\Rightarrow \mathbf{N} \subseteq Dom(f)$ (Definition of $\subseteq$) . . . (II)

Hence, $Dom(f) \subseteq \mathbf{N}$ and $\mathbf{N} \subseteq Dom(f)$ ((I),(II),I6)

$\Rightarrow Dom(f) = \mathbf{N}$. (Theorem 3.2.3(ii))

(ii) Let $(x, y) \in f$ and $(x, z) \in f$.

Then, $y = 2x$ and $z = 2x$ (Definition of f)

$\Rightarrow y = z$. ($\text{Sub}_=$)

Hence, f is a function from $\mathbf{N}$ to $\mathbf{E}$. ■

In conventional notation, a function is usually defined by specifying its domain and the image of each element in the domain rather than as a set of ordered pairs. Specifically, a function is usually defined in the following way: "Let $f : X \to Y$ be such that $f(x) = \cdots, \forall x \in X$", where the ellipsis . . . is an expression or a statement describing how the image of x under f, denoted $f(x)$, is determined.

Using the conventional notations, the definition of f above can be rewritten as $f : \mathbf{N} \to \mathbf{E}$ such that $f(x) = 2x, \forall x \in \mathbf{N}$.

(v) (*The identity functions*) Let X be any set. The *relation of equality in* X (i.e $X_= = \{(x, y) \in X \times X \mid x = y\}$) is a function from X to X.

Proof: (i) Since $X_=$ is a relation in X, $Dom(X_=) \subseteq X$.

On the other hand, if $x \in X$, then $(x = x) \Rightarrow (x, x) \in X_=$. (Definition of $X_=$)

$\Rightarrow x \in Dom(X_=)$. (Definition of *domain*)

Consequently, $X \subseteq Dom(X_=)$. (Definition of $\subseteq$)

Hence, $Dom(X_=) = X$. (Theorem 3.2.3(ii))

(ii) Let $(x, y) \in X_=$ and $(x, z) \in X_=$.

Then, $y = x$ and $z = x$ (Definition of $X_=$)

It follows that $y = z$. (= is transitive)

Hence, $X_=$ is a function from X to X. ■

Owing to the above result, the relation of equality in X is also called the ***identity function*** in X and in the conventional terminology of function, it is defined as $I_X : X \to X$ such that $I_X(x) = x, \forall x \in X$.

(vi) (The *constant functions*) Let X, Y be any two sets and $a \in Y$. The relation $C_a = \{(x, a) \mid x \in X\}$ is a function from X to Y. The proof is similar to those of (iv) and (v) above and is left as an exercise. The function C_a maps every element in X to the same element a in Y. It is thus called a ***constant function*** and can be redefined as: $C_a : X \to Y$ such that $C_a(x) = a, \forall x \in X$.

(vii) (The *addition function*) Let $\mathbf{R}$ be the set of real numbers. The addition operator, +, can be defined as a function as follows:

$$add : \mathbf{R} \times \mathbf{R} \to \mathbf{R} \text{ such that } add((a, b)) = a + b, \forall (a, b) \in \mathbf{R} \times \mathbf{R}.$$

Lemma 6.1.1: *Let $f : X \to Y$ and $g : X \to Y$. Then, $f = g$ iff $f(x) = g(x), \forall x \in X$.*

Proof: (Direct proof) Let $f : X \to Y$ *and* $g : X \to Y$.
(Bidirectional proof)
$\Rightarrow$) (Direct Proof) Suppose $f = g$. $\cdots$ (I)
Let $x \in X$. Then $x \in Dom(f)$ $\quad (Dom(f) = X)$
$\Rightarrow (\exists y \in Y)(x, y) \in f$ $\quad$ (Definition of $Dom(f)$)
$\Rightarrow (x, a) \in f$ $\quad$ (EI)
$\Rightarrow (x, a) \in g$ $\quad$ ((I), Sub$_=$)
$\Rightarrow (x, a) \in f \wedge (x, a) \in g$ $\quad$ (I6)
$\Rightarrow (f(x) = a) \wedge (g(x) = a)$ $\quad$ (equivalent notations)
$\Rightarrow f(x) = g(x)$. $\quad$ (Sub$_=$)

Hence, $x \in X \Rightarrow f(x) = g(x)$
$\Rightarrow (\forall x)(x \in X \Rightarrow f(x) = g(x))$ $\quad$ (Gen)

$\Leftarrow$) (Direct Proof) Suppose $f(x) = g(x), \forall x \in X$.

$$\begin{aligned} (x, y) \in f &\Leftrightarrow y = f(x) && (equivalent\ notations) \\ &\Leftrightarrow y = g(x) && (f(x) = g(x), \text{Sub}_=) \\ &\Leftrightarrow (x, y) \in g && (equivalent\ notations) \end{aligned}$$

Hence, $(x, y) \in f \Leftrightarrow (x, y) \in g$
$\Rightarrow (\forall z)(z \in f \Leftrightarrow z \in g)$ $\quad$ (Gen)
$\Rightarrow f = g$. $\quad$ (Principle of Extension) ■

The definition of the image of an element in the domain can be naturally generalized to a subset of the domain as follows.

Definition: Let $f : X \to Y$ and $A \subseteq X$. The ***image*** of A under f is the set:
$f(A) = \{y \in Y \mid (\exists x \in A)(x, y) \in f\} = \{y \in Y \mid (\exists x)(x \in A \wedge y = f(x))\} = \{f(x) \in Y \mid x \in A\}$.

Example 6.1.19
(i) Let $f : X \to Y$. $f(X) = \{y \in Y \mid (\exists x \in X)(x, y) \in f\} = Ran(f)$.
(ii) Let f be defined as in Example 6.1.18(iv). Let $A = \{x \in \mathbf{N} \mid (\exists k \in \mathbf{N})x = 2k\}$.

Then, $f(A) = \{y \in \mathbf{E} \mid (\exists x \in A)(x, y) \in f\}$ (Definition of $f(A)$)
$= \{y \in \mathbf{E} \mid (\exists x)(x \in A \wedge (x, y) \in f)\}$ (equivalent notations)
$= \{y \in \mathbf{E} \mid (\exists x)((\exists k \in \mathbf{N})(x = 2k) \wedge (x, y) \in f)\}$ (Definition of A)
$= \{y \in \mathbf{E} \mid (\exists x)(\exists k \in \mathbf{N})(x = 2k \wedge y = f(x))\}$ (FE6, equivalent notations)
$= \{y \in \mathbf{E} \mid (\exists k \in \mathbf{N})y = f(2k)\}$ ($\text{Sub}_=$, I2)
$= \{y \in \mathbf{E} \mid (\exists k \in \mathbf{N})y = 2(2k)\}$ (Definition of f)
$= \{y \in \mathbf{E} \mid (\exists k \in \mathbf{N})y = 4k\}$.

Therefore, $f(A)$ consists of all the positive multiples of 4. ■

Definition: Let f, g be two functions such that $g \subseteq f$. Then g is a ***restriction of*** f and f is an ***extension of*** g. Moreover, let $f : X \to Y$ and $Z \subseteq X$ such that $Dom(g) = Z$. Then g is ***restriction of*** f ***to*** $\mathbf{Z}$ and f is an ***extension of*** g ***to*** $\mathbf{X}$. The function g is customary denoted by $f \mid Z$, and $f \mid Z = \{(x, y) \in f \mid x \in Z\}$.

Example 6.1.20 Consider the functions in Example 6.1.18. Since $f_1 \subseteq f_3$, f_3 is an extension of f_1 to $\{a, b, c, d\}$, and f_1 is a restriction of f_3 to $\{a, b, c\}$.

Definition: The set of all functions from X to Y is the set: $Y^X = \{f \mid f : X \to Y\}$.

Note that Y^X is a subset of the power set $\mathcal{P}(X \times Y)$.

Example:

1. Let $X = \{a, b, c\}$ and $Y = \{x, y\}$. Then $Y^X = \{f_1, f_2, f_3, f_4, f_5, f_6, f_7, f_8\}$, where
$f_1 = \{(a, x), (b, x), (c, x)\}$
$f_2 = \{(a, y), (b, x), (c, x)\}$
$f_3 = \{(a, x), (b, y), (c, x)\}$
$f_4 = \{(a, x), (b, x), (c, y)\}$
$f_5 = \{(a, y), (b, y), (c, x)\}$
$f_6 = \{(a, y), (b, x), (c, y)\}$
$f_7 = \{(a, x), (b, y), (c, y)\}$
$f_8 = \{(a, y), (b, y), (c, y)\}$.

Note that X has 3 elements, Y has 2 elements, and the total number of functions from X to Y is $2^3 = 8$

In general, let X have m elements and Y have n elements. Since for each $x \in X$, there are n different ways of assigning an image to x, there are altogether $n \cdot n \cdots n = n^m$ ways of assigning images to all the m elements in X, resulting in n^m functions from X to Y.

2. $Y^\emptyset = \{\emptyset\}$.

By Example 6.1.18(ii), $\emptyset \in Y^\emptyset$. i.e. $\{\emptyset\} \subseteq Y^\emptyset$. . . . (I)

Let $f \in Y^\emptyset$. Then $f : \emptyset \to Y$.

Suppose $f \neq \emptyset$. Then $\exists (x, y), (x, y) \in f$ (Lemma 3.5.8)
$\Rightarrow (a, b) \in f$ (EI)
$\Rightarrow a \in \emptyset \wedge b \in Y$ (Definition of f)
$\Rightarrow a \in \emptyset$ (I2)
$\Rightarrow (\exists x)x \in \emptyset$ (EQ)
$\Rightarrow \emptyset \neq \emptyset$, a contradiction! (Lemma 3.5.8)

Therefore, $f = \emptyset \Rightarrow f \in \{\emptyset\}$.

We thus have: $f \in Y^{\emptyset} \Rightarrow f \in \{\emptyset\}$

$\Rightarrow (\forall f)(f \in Y^{\emptyset} \Rightarrow f \in \{\emptyset\})$ (Gen)

$\Rightarrow Y^{\emptyset} \subseteq \{\emptyset\}$. (Definition of $\subseteq$) . . . (II)

Hence, $(\{\emptyset\} \subseteq Y^{\emptyset}) \wedge (Y^{\emptyset} \subseteq \{\emptyset\})$ ((I),(II), I6)

$\Rightarrow Y^{\emptyset} = \{\emptyset\}$. (3.2.3(ii)) ■

Let $f : X \rightarrow Y$ and $X = U \cup V$ such that $U \cap V = \emptyset$. Let $g : U \rightarrow Y$ ($h : V \rightarrow Y$, respectively) be the function such that the value of $f(x)$, $\forall x \in U$ ($\forall x \in V$, respectively) is determined by $g(x)$ ($h(x)$, respectively). The function f is customary defined as:

$$f(x) = \begin{cases} g(x) & if\ x \in U \\ h(x) & if\ x \in V \end{cases}$$

Let $\alpha_{U}(x)$ and $\alpha_{V}(x)$ be the defining properties of the set U and V, respectively. Then, the function f may also be defined as:

$$f(x) = \begin{cases} g(x) & if\ \alpha_{U}(x) \\ h(x) & if\ \alpha_{V}(x) \end{cases}$$

Example 6.1.21

(i) Let $\mathbf{N}$ be the set of all positive integers, $\mathbf{Z}$ be the set of all integers, and $f : \mathbf{N} \rightarrow \mathbf{Z}$ such that $f = \{(x, y) \in \mathbf{N} \times \mathbf{Z} \mid (x \in \mathbf{E} \wedge y = 2x) \vee (x \in \mathbf{O} \wedge y = -2x)\}$, where $\mathbf{E} = \{x \in \mathbf{N} \mid x\ is\ even\}$ and $\mathbf{O} = \{x \in \mathbf{N} \mid x\ is\ odd\}$. (i.e. $f(x) = 2x$ if x is an even positive integer, and $f(x) = -2x$ if x is an odd positive integer.)

It is easily verified that $\mathbf{E} \cup \mathbf{O} = \mathbf{N}$ and $\mathbf{E} \cap \mathbf{O} = \emptyset$. (Verify!)

Let $g : \mathbf{E} \rightarrow \mathbf{Z}$ such that $g(x) = 2x, \forall x \in \mathbf{E}$, and

$h : \mathbf{O} \rightarrow \mathbf{Z}$ such that $h(x) = -2x, \forall x \in \mathbf{O}$.

Then g defines the value of $f(x)$, $\forall x \in \mathbf{E}$, and h defines the value of $f(x)$, $\forall x \in \mathbf{O}$.

As a result, f can be defined as

$$f(x) = \begin{cases} 2x & if\ x \in \mathbf{E} \\ -2x & if\ x \in \mathbf{O} \end{cases}$$

or

$$f(x) = \begin{cases} 2x & if\ x\ is\ even \\ -2x & if\ x\ is\ odd \end{cases}$$

■

6.2 Properties of Functions

Definition: A function $f : X \rightarrow Y$ is *onto* or *surjective* (denoted by $f : X \overset{onto}{\rightarrow} Y$) iff $(\forall y \in Y)(\exists x \in X)(x, y) \in f$, or equivalently, $(\forall y \in Y)(\exists x \in X)y = f(x)$.

Definition: A function $f : X \rightarrow Y$ is *one-to-one* or *injective* (denoted by $f : X \overset{1-1}{\rightarrow} Y$) iff $(x, z) \in f \wedge (y, z) \in f \Rightarrow x = y$, or equivalently, $f(x) = f(y) \Rightarrow x = y$.

Definition: A function $f : X \rightarrow Y$ is a *one-to-one correspondence* or *bijective* (denoted by $f : X \underset{onto}{\overset{1-1}{\rightarrow}} Y$) iff it is both one-to-one and onto.

Remark: In the definition of *onto* function, $(\forall y \in Y)(\exists x \in X)y = f(x)$ is equivalent to $(\forall y)(y \in Y \Rightarrow ((\exists x)x \in X \wedge y = f(x)))$.

Example 6.2.22

(i) In Example 6.1.18(i), the function f_2 from X to Y is neither one-to-one nor onto. This is because $(a, x) \in f_2 \wedge (c, x) \in f_2$ but $a \neq c$, and $z \in Y$ but $\sim (\exists x \in X)(x, z) \in f_2$. the function f_3 from X to Y is *not* one-to-one but is onto. The function f_4 from X to Z is both one-to-one and onto.

(ii) In Example 6.1.18(iv), the function $f : \mathbf{N} \rightarrow \mathbf{E}$ such that $f(x) = 2x, \forall x \in \mathbf{N}$ is one-to-one and onto.

Proof: Let $x, y \in \mathbf{N}$ be such that $f(x) = f(y)$.

Then, $2x = 2y$ (Definition of f)

$\Rightarrow x = y$

Therefore, f is one-to-one.

Let $y \in \mathbf{E}$. Then $(\exists k)(k \in \mathbf{N} \wedge y = 2k)$ (definition of $\mathbf{E}$)

$\Rightarrow (\exists k)(k \in \mathbf{N} \wedge y = f(k))$ (Definition of f)

Therefore, f is onto. □

(iii) In Example 6.1.18(v), the identity function I_X is a one-to-one correspondence.

(iv) In Example 6.1.18(vi), the constants function C_a is not one-to-one unless X is a singleton. It is not onto unless $Y = \{a\}$.

(v) In Example 6.1.18(viii), the function *add* is not one-to-one because $add((1, 3)) = add((3, 1)) = 4$ but $(1, 3) \neq (3, 1)$. It is, however, onto. This is because $\forall y \in \mathbf{R}, \exists (y, 0) \in \mathbf{R} \times \mathbf{R}, ((y, 0), y) \in add$.
■

6.3 Inverse Functions and Composite Functions

Since functions are relations, all the operators defined for relations are applicable to functions. However, we are particularly interested in the inverse and composite operators only.

Definition: Let $f : A \rightarrow B$. The ***inverse*** of f is the relation f^{-1} from B to A such that $f^{-1} = \{(y, x) \mid (x, y) \in f\}$.

Although f is a function, f^{-1} may not be a function as shown in the following examples.

Example 6.3.23

(i) Let $f : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3\}$ be such that $f = \{(1, 2), (4, 3), (2, 3), (3, 1)\}$. Then $f^{-1} = \{(2, 1), (3, 4), (3, 2), (1, 3)\}$ is not a function because $(3, 2) \in f^{-1}$ and $(3, 4) \in f^{-1}$ but $2 \neq 4$.

(ii) Let $\mathbf{Z}$ be the set of all integers. Define $f : \mathbf{Z} \rightarrow \mathbf{Z}$ such that $f(x) = 2x, \forall x \in \mathbf{N}$. Then $f^{-1} = \{(x, y) \in \mathbf{Z} \times \mathbf{Z} \mid x = 2y\}$ is not a function from $\mathbf{Z}$ to $\mathbf{Z}$. This is because when $x = 1$, $x \in \mathbf{Z}$ but $\sim (\exists y \in \mathbf{Z})(x, y) \in f^{-1}$

(iii) Let $\mathbf{Z}$ be the set of integers and $\mathbf{E}$ be the set of even inetgers. Define $f : \mathbf{Z} \rightarrow \mathbf{E}$ such that $f(x) = 2x, \forall x \in \mathbf{N}$. Then $f^{-1} = \{(x, y) \in \mathbf{E} \times \mathbf{Z} \mid x = 2y\}$ is a function from $\mathbf{E}$ to $\mathbf{Z}$.

(iv) The inverse of the *square* function for real numbers is the *square root* function for real numbers.
■

We shall further generalize the concept of *the image of a set under a function* to a relation.

Definition: Let $R \subseteq X \times Y$ be a relation from X to Y and $A \subseteq X$. The ***image*** of A under R is the set:

$$R(A) = \{y \in Y | (\exists x \in A)(x, y) \in R\}.$$

Definition: Let $f : X \to Y$. Then $f^{-1} \subseteq Y \times X$ is a relation from Y to X. Let $B \subseteq Y$. The ***inverse image*** of B under f is the image of B under f^{-1} which is the set:

$$f^{-1}(B) = \{x \in X | (\exists y \in B)(y, x) \in f^{-1}\} = \{x \in X | (\exists y \in B)(x, y) \in f\} = \{x \in X \mid f(x) \in B\}.$$

Example: Consider the function $add : \mathbf{R} \times \mathbf{R} \to \mathbf{R}$. Let $B = \{0\} \subseteq \mathbf{R}$.

$$\begin{aligned} add^{-1}(\{0\}) &= \{(x, y) \in \mathbf{R} \times \mathbf{R} | add((x, y)) \in \{0\}\} \\ &= \{(x, y) \in \mathbf{R} \times \mathbf{R} | x + y = 0\} \\ &= \{(x, y) \in \mathbf{R} \times \mathbf{R} | y = -x\}. \\ &= \{(x, -x) \in \mathbf{R} \times \mathbf{R} | x \in \mathbf{R}\}. \quad \blacksquare \end{aligned}$$

The following lemma gives a sufficient and necessary condition that a function must satisfy in order for its inverse to be a function.

Lemma 6.3.1: *Let $f : A \to B$, Then f^{-1} is a function from B to A iff f is one-to-one and onto.*

Proof: (Direct proof) Let $f : A \to B$.

(Bidirectional proof)

$\Rightarrow$) (Direct proof) Suppose f^{-1} is a function from B to A.

Then $Dom(f^{-1}) = B$ (Definition of *function*)

$\Rightarrow (\forall y \in B)(\exists x \in A)(y, x) \in f^{-1}$ (Definition of *domain*)

$\Rightarrow (\forall y \in B)(\exists x \in A)(x, y) \in f$ (Definition of f^{-1})

i.e. f is onto.

$(x_1, y) \in f \wedge (x_2, y) \in f$

$\Rightarrow (y, x_1) \in f^{-1} \wedge (y, x_2) \in f^{-1}$ (Definition of f^{-1})

$\Rightarrow x_1 = x_2$ (Definition of *function*)

i.e. f is one-to-one.

$\Leftarrow$) (Direct proof) Suppose f is one-to-one and onto.

Let $y \in B$. Then $(\exists x \in A)(x, y) \in f$ (f is onto)

$\Rightarrow (\exists x \in A)(y, x) \in f^{-1}$ (Definition of f^{-1})

$\Rightarrow y \in Dom(f^{-1})$ (Definition of *domain*)

Therefore, $B \subseteq Dom(f^{-1})$. $\cdots$(I)

But, by Lemma 5.1.1, $Dom(f^{-1}) \subseteq B$. $\cdots$(II)

We thus have $(B \subseteq Dom(f^{-1})) \wedge (Dom(f^{-1}) \subseteq B)$ ((I),(II), I6)

$\Rightarrow Dom(f^{-1}) = B$. (Theorem 3.2.3(ii))

Suppose $(x, y_1) \in f^{-1} \wedge (x, y_2) \in f^{-1}$.

Then $(y_1, x) \in f \wedge (y_2, x) \in f$. (Definition of f^{-1})

$\Rightarrow y_1 = y_2$ (f is one-to-one)

Therefore, f^{-1} is a function from B to A. $\blacksquare$

Theorem 6.3.2: *$f : A \underset{onto}{\overset{1-1}{\rightarrow}} B$ if and only if $f^{-1} : B \underset{onto}{\overset{1-1}{\rightarrow}} A$.*

Proof: (Bidirectional proof)

$\Rightarrow$) (Direct proof) Let $f : A \underset{onto}{\overset{1-1}{\rightarrow}} B$. Then $\left(f^{-1}\right)^{-1} : A \rightarrow B$. (Theorem 5.6.5(i))

$\Rightarrow \left(f^{-1}\right) : B \underset{onto}{\overset{1-1}{\rightarrow}} A$. (Lemma 6.3.1)

$\Leftarrow$) Let $f^{-1} : B \underset{onto}{\overset{1-1}{\rightarrow}} A$. Then $f : A \underset{onto}{\overset{1-1}{\rightarrow}} B$ (Lemma 6.3.1) ■

Since functions are relations, it makes sense to talk about the composition of functions. Although we have shown above that the inverse of a function may not be a function, it turns out that the composite of two functions is a function. This is verified as follows.

Let $f : A \rightarrow B$ and $g : B \rightarrow C$. we want to prove that the relation $f \circ g$ is a function from A to C.

(i) By Lemma 5.1.1, $Dom(f \circ g) \subseteq A$. . . . (A)

It remains to prove that $A \subseteq Dom(f \circ g)$.

First, we shall prove that $x \in A \Rightarrow x \in Dom(f \circ g)$.

(Direct proof) Let $x \in A$. . . . (I)

Then $f : A \rightarrow B \Rightarrow Dom(f) = A$ (Definition of function)

$\Rightarrow x \in Dom(f)$ ((I), Sub$_=$)

$\Rightarrow (\exists y \in B)(x, y) \in f$ (Definition of $Dom(f)$)

$\Rightarrow (b \in B) \wedge (x, b) \in f$ (EI)

$\Rightarrow b \in B$ and $(x, b) \in f$ (E9,I2) . . . (II)

$g : B \rightarrow C \Rightarrow Dom(g) = B$ (Definition of function)

$\Rightarrow b \in Dom(g)$ ((II), Sub$_=$)

$\Rightarrow (\exists z \in C)(b, z) \in g$ (Definition of $Dom(g)$)

$\Rightarrow (c \in C) \wedge (b, c) \in g$ (EI)

$\Rightarrow c \in C$ and $(b, c) \in g$ (E9,I2) . . . (III)

It follows that $b \in B \wedge (x, b) \in f \wedge (b, c) \in g$ ((II),(III),I6)

$\Rightarrow (\exists y)(y \in B \wedge (x, y) \in f \wedge (y, c) \in g)$ (EQ)

$\Rightarrow (x, c) \in f \circ g$ (Definition of $\circ$)

$\Rightarrow x \in Dom(f \circ g)$ (Definition of $Dom(f \circ g)$)

Hence, $x \in A \Rightarrow x \in Dom(f \circ g)$

$\Rightarrow (\forall x)(x \in A \Rightarrow x \in Dom(f \circ g))$ (Gen)

$\Rightarrow A \subseteq Dom(f \circ g)$. (Definition of $\subseteq$) . . . (B)

Thus, $Dom(f \circ g) \subseteq A \wedge A \subseteq Dom(f \circ g)$. ((A),(B),I6)

$\Rightarrow Dom(f \circ g) = A$. (Theorem 3.2.3(ii))

(ii) Let $(x, z_1) \in f \circ g \wedge (x, z_2) \in f \circ g$. Then,

$(\exists y_1 \in B)((x, y_1) \in f \wedge (y_1, z_1) \in g) \wedge (\exists y_2 \in B)((x, y_2) \in f \wedge (y_2, z_2) \in g)$ (Definition of $\circ$)

$\Rightarrow (\exists y_1 \in B)((x, y_1) \in f \wedge (y_1, z_1) \in g)$ and $(\exists y_2 \in B)((x, y_2) \in f \wedge (y_2, z_2) \in g)$ (E9,I2)

$\Rightarrow (b_1 \in B \wedge (x, b_1) \in f \wedge (b_1, z_1) \in g)$ and $(b_2 \in B \wedge (x, b_2) \in f \wedge (b_2, z_2) \in g)$ (EI)

$\Rightarrow b_1 \in B$ and $(x, b_1) \in f$ and $(b_1, z_1) \in g$ and $b_2 \in B$ and $(x, b_2) \in f$ and $(b_2, z_2) \in g$ (E9,I2)

$\Rightarrow ((x, b_1) \in f \wedge (x, b_2) \in f)$ and $((b_1, z_1) \in g \wedge (b_2, z_2) \in g)$ (I6)

$\Rightarrow b_1 = b_2$ and $((b_1, z_1) \in g \wedge (b_2, z_2) \in g)$ (f is a function)

$\Rightarrow ((b_1, z_1) \in g \wedge (b_1, z_2) \in g)$ ($\text{Sub}_=$)

$\Rightarrow z_1 = z_2$. (g is a function)

Hence, $f \circ g$ is a function from A to C which leads us to the following definition.

Definition: Let $f : A \to B$ and $g : B \to C$. The ***composition*** of f and g is the function $f \circ g : A \to C$.

[**Remark:** $(f \circ g)(x) = g(f(x)), \forall x \in A$.

Proof: Let $x \in A$.

Then, $y = (f \circ g)(x) \Rightarrow (x, y) \in f \circ g$ (Equivalent notation)

$\Rightarrow (\exists z \in B)((x, z) \in f \wedge (z, y) \in g)$ (Definition of $\circ$)

$\Rightarrow (\exists z \in B)(z = f(x) \wedge (z, y) \in g)$ (Equivalent notation)

$\Rightarrow a \in B \wedge a = f(x) \wedge (a, y) \in g$ (EI)

$\Rightarrow a = f(x)$ and $(a, y) \in g$ (I2,E9)

$\Rightarrow (f(x), y) \in g$ ($\text{Sub}_=$)

$\Rightarrow y = g(f(x))$ (Equivalent notation)

$\Rightarrow (f \circ g)(x) = g(f(x))$ ($\text{Sub}_=$)

Hence, $(f \circ g)(x) = g(f(x)), \forall x \in A$.]

Lemma 6.3.3: Let $f : A \overset{1-1}{\to} B$ and $g : B \overset{1-1}{\to} C$. Then $f \circ g : A \overset{1-1}{\to} C$.

Proof: (Direct proof) Let $f : A \overset{1-1}{\to} B$ and $g : B \overset{1-1}{\to} C$.

We want to prove that$(f \circ g)(x_1) = (f \circ g)(x_2) \Rightarrow x_1 = x_2$.

(Direct proof)

Let $(f \circ g)(x_1) = (f \circ g)(x_2)$.

Then $g(f(x_1)) = g(f(x_2))$ (equivalent notations) . . . (I)

But $g : B \overset{1-1}{\to} C$

$\Rightarrow (g(f(x_1)) = g(f(x_2)) \Rightarrow f(x_1) = f(x_2))$ (g is one-to-one) . . . (II)

$\Rightarrow f(x_1) = f(x_2)$ ((I),(II), I3) . . . (III)

But $f : A \overset{1-1}{\to} B$

$\Rightarrow (f(x_1) = f(x_2) \Rightarrow x_1 = x_2)$ (f is one-to-one) . . . (IV)

$\Rightarrow x_1 = x_2$ ((III),(IV), I3) ■

Lemma 6.3.4: Let $f : A \overset{onto}{\to} B$ and $g : B \overset{onto}{\to} C$. Then $f \circ g : A \overset{onto}{\to} C$.

Proof: (Direct Proof) Let $f : A \overset{onto}{\to} B$ and $g : B \overset{onto}{\to} C$.

We want to prove that $(\forall z \in C)(\exists x \in A)(x, z) \in f \circ g$, or equivalently,

$$(\forall z)(z \in C \Rightarrow (\exists x \in A)(x, z) \in f \circ g)$$

(Direct Proof)

Let $z \in C$. . . . (I)

Since $g : B \overset{onto}{\to} C$,

therefore, $(\forall w \in C)(\exists y \in B)(y, w) \in g$ (Definition of onto finction)

$\Rightarrow (z \in C \Rightarrow (\exists y \in B)(y, z) \in g)$ (UI) . . . (II)

$\Rightarrow (\exists y \in B)(y, z) \in g$ (I),(II), I3

$\Rightarrow b \in B \wedge (b, z) \in g$ (EI)

$\Rightarrow b \in B$ and $(b, z) \in g$ (E9, I2) . . . (III)

Since $f : A \overset{onto}{\rightarrow} B$,

therefore, $(\forall y \in B)(\exists x \in A)(x, y) \in f$ (Definition of onto finction)

$\Rightarrow (b \in B \Rightarrow (\exists x \in A)(x, b) \in f)$ (UI) . . . (IV)

$\Rightarrow (\exists x \in A)(x, b) \in f$ (III),(IV),I3)

$\Rightarrow a \in A \wedge (a, b) \in f$ (EI)

$\Rightarrow a \in A$ and $(a, b) \in f$ (E9, I2) . . . (V)

$\Rightarrow b \in B \wedge (a, b) \in f \wedge (b, z) \in g$ ((V),(III), I6)

$\Rightarrow (\exists y \in B)((a, y) \in f \wedge (y, z) \in g)$ (EQ)

$\Rightarrow (a, z) \in f \circ g$ (Definition of $\circ$) . . . (VI)

$\Rightarrow (a \in A) \wedge (a, z) \in f \circ g$ ((V),(VI), I6)

$\Rightarrow (\exists x \in A)(x, z) \in f \circ g$ (EQ)

Hence, $z \in C \Rightarrow (\exists x \in A)(x, z) \in f \circ g$

$\Rightarrow (\forall z \in C)(\exists x \in A)(x, z) \in f \circ g$ (Gen)

$\Rightarrow f \circ g : A \overset{onto}{\rightarrow} C.$ (Definition of ontoness) ■

Combining the above two lemmas, we obtain:

Theorem 6.3.5: Let $f : A \underset{onto}{\overset{1-1}{\rightarrow}} B$ and $g : B \underset{onto}{\overset{1-1}{\rightarrow}} C$. Then $f \circ g : A \underset{onto}{\overset{1-1}{\rightarrow}} C$.

6.4 Examples

1. Let $f : X \rightarrow Y$. Prove that $\forall A, B \subseteq X, f(A \cup B) = f(A) \cup f(B)$.

Solution: $y \in f(A \cup B) \Leftrightarrow (\exists x \in A \cup B) y = f(x)$ (Definition of $f(A \cup B)$)

$\Leftrightarrow (\exists x)(x \in (A \cup B) \wedge y = f(x))$ (equivalent notations)

$\Leftrightarrow (\exists x)((x \in A \vee x \in B) \wedge y = f(x))$ (Definition of $\cup$)

$\Leftrightarrow (\exists x)((x \in A \wedge y = f(x)) \vee (x \in B \wedge y = f(x)))$ (E10, E14)

$\Leftrightarrow (\exists x)(x \in A \wedge y = f(x)) \vee (\exists x)(x \in B \wedge y = f(x))$ (FE10)

$\Leftrightarrow (\exists x \in A)(y = f(x)) \vee (\exists x \in B)(y = f(x))$ (equivalent notations)

$\Leftrightarrow y \in f(A) \vee y \in f(B)$ (Definitions of $f(A), f(B)$)

$\Leftrightarrow y \in f(A) \cup f(B)$ (Definition of $\cup$) ■

2. Let $f : X \rightarrow Y$. Prove that $\forall A, B \subseteq X, f(A \cap B) \subseteq f(A) \cap f(B)$.

Solution: $y \in f(A \cap B) \Rightarrow (\exists x \in A \cap B) y = f(x)$ (Definition of $f(A \cap B)$)

$\Rightarrow (\exists x)(x \in (A \cap B) \wedge y = f(x))$ (equivalent notations)

$\Rightarrow (\exists x)((x \in A \wedge x \in B) \wedge y = f(x))$ (Definition of $\cap$)

$\Rightarrow (\exists x)((x \in A \wedge x \in B) \wedge y = f(x))$ (Definition of $\cap$)

$\Rightarrow (\exists x)((x \in A \wedge x \in B) \wedge (y = f(x)) \wedge (y = f(x)))$ (E3)

$\Rightarrow (\exists x)((x \in A \wedge y = f(x)) \wedge (x \in B \wedge y = f(x)))$ (E9,E11)

$\Rightarrow (\exists x)(x \in A \wedge y = f(x)) \wedge (\exists x)(x \in B \wedge y = f(x))$ (Examples 2.8(3))

$\Rightarrow (\exists x \in A)(y = f(x)) \wedge (\exists x \in B)(y = f(x))$ (equivalent notations)

$\Rightarrow y \in f(A) \wedge y \in f(B)$ (Definitions of $f(A), f(B)$)
$\Rightarrow y \in f(A) \cap f(B)$ (Definition of $\cap$) ■

3. Let $f : X \to Y$. Prove that $\forall A, B \subseteq Y, f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$.

Solution: $x \in f^{-1}(A \cup B) \Leftrightarrow f(x) \in (A \cup B)$ (Definition of $f^{-1}(A \cup B)$)
$\Leftrightarrow f(x) \in A \vee f(x) \in B$ (Definition of $\cup$)
$\Leftrightarrow x \in f^{-1}(A) \vee x \in f^{-1}(B)$ (Definitions of $f^{-1}(A), f^{-1}(B)$)
$\Leftrightarrow x \in f^{-1}(A) \cup f^{-1}(B)$ (Definition of $\cup$) ■

4. Let $f : X \to Y$. Prove that $\forall A, B \subseteq Y, f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$.

Solution: $x \in f^{-1}(A \cap B) \Leftrightarrow f(x) \in (A \cap B)$ (Definitions of $f^{-1}(A \cap B)$)
$\Leftrightarrow f(x) \in A \wedge f(x) \in B$ (Definition of $\cap$)
$\Leftrightarrow x \in f^{-1}(A) \wedge x \in f^{-1}(B)$ (Definitions of $f^{-1}(A), f^{-1}(B)$)
$\Leftrightarrow x \in f^{-1}(A) \cap f^{-1}(B)$ (Definition of $\cap$) ■

6.5 The Characteristic Functions

Definition: Let **U** be the universal set, and $A \subseteq \mathbf{U}$. The ***characteristic function*** of A is the function: $\chi_A : \mathbf{U} \to \{0, 1\}$ such that

$$\chi_A(x) = \begin{cases} 0 & if\ x \in \overline{A} \\ 1 & if\ x \in A \end{cases}$$

Note that $x \in A$ iff $\chi_A(x) = 1$.

The following are some examples of characteristic functions.

Let **U=N** be the set of positive integers.

1. Let $A = \emptyset$. Then $\chi_A = C_0$.
2. Let $A = \mathbf{N}$. Then $\chi_A = C_1$.
3. Let **E** be the set of all even inetgers. Then $\chi_{\mathbf{E}}(x) = \begin{cases} 0 & if\ x\ is\ odd \\ 1 & if\ x\ is\ even \end{cases}$.

The characteristic function χ_A from **U** to {0,1} is not one-to-one unless **U** is an unordered pairs and A is a singleton, or **U** is a singleton. χ_A is onto so long as $A \neq \mathbf{U}$.

Lemma 6.5.1: Let A and B be sets.

(i) $\forall x \in \mathbf{U}, \chi_A(x) = 0\ iff\ A = \emptyset$.
(ii) $\forall x \in \mathbf{U}, \chi_A(x) = 1\ iff\ A = \mathbf{U}$.
(iii) $\forall x \in \mathbf{U}, \chi_A(x) \leq \chi_B(x)\ iff\ A \subseteq B$.
(iv) $\forall x \in \mathbf{U}, \chi_A(x) = \chi_B(x)\ iff\ A = B$.
(v) $\forall x \in \mathbf{U}, \chi_{A \cup B}(x) = \chi_A(x) + \chi_B(x) - \chi_A(x) \cdot \chi_B(x)$.
(vi) $\forall x \in \mathbf{U}, \chi_{A \cap B}(x) = \chi_A(x) \cdot \chi_B(x)$.

Proof:

(i) $\forall x \in \mathbf{U}, \chi_A(x) = 0 \Leftrightarrow \forall x \in \mathbf{U}, x \notin A$ (Definition of χ_A)
$\Leftrightarrow (\forall x)(x \in \mathbf{U} \Rightarrow x \notin A)$ (equivalent notation)
$\Leftrightarrow (\forall x)(\sim x \in \mathbf{U} \vee x \notin A)$ (E18)
$\Leftrightarrow (\forall x)(\sim true \vee x \notin A)$ (Definition of **U**)
$\Leftrightarrow (\forall x)(false \vee x \notin A)$ (Theorem 1.7.2(i))

$\Leftrightarrow (\forall x)(x \notin A)$ (E10, E6)
$\Leftrightarrow \sim (\exists x)(x \in A)$ (FE8)
$\Leftrightarrow A = \emptyset$ (Lemma 3.5.8, E15)

(ii) $\forall x \in \mathbf{U}, \chi_A(x) = 1 \Leftrightarrow \forall x \in \mathbf{U}, x \in A$ (Definition of χ_A)
$\Leftrightarrow \forall x \in \mathbf{U}, x \notin \overline{A}$ (Lemma 4.3.5)
$\Leftrightarrow \forall x \in \mathbf{U}, \chi_{\overline{A}}(x) = 0$ (Definition of $\chi_{\overline{A}}$)
$\Leftrightarrow \overline{A} = \emptyset$ (Lemma 6.5.1(i))
$\Leftrightarrow \overline{A} = \overline{\mathbf{U}}$ (Theorem 4.3.7(ii))
$\Leftrightarrow (\overline{A} \subseteq \overline{\mathbf{U}}) \wedge (\overline{\mathbf{U}} \subseteq \overline{A})$ ($\Rightarrow$) Theorem 3.2.4; $\Leftarrow$) Theorem 3.2.3(ii))
$\Leftrightarrow \left(\overline{\overline{\mathbf{U}}} \subseteq \overline{\overline{A}}\right) \wedge \left(\overline{\overline{A}} \subseteq \overline{\overline{\mathbf{U}}}\right)$ (Theorem 4.3.7(iv))
$\Leftrightarrow (\mathbf{U} \subseteq A) \wedge (A \subseteq \mathbf{U})$ (Theorem 4.3.7(i))
$\Leftrightarrow A = \mathbf{U}.$ ($\Rightarrow$) Theorem 3.2.3(ii); $\Leftarrow$) Theorem 3.2.4)

(iii) $\forall x \in \mathbf{U}, \chi_A(x) \leq \chi_B(x) \Leftrightarrow \forall x \in \mathbf{U}, ((\chi_A(x) = 0) \vee (\chi_A(x) = 1 \wedge \chi_B(x) = 1))$
$\Leftrightarrow \forall x \in \mathbf{U}, ((x \notin A) \vee (x \in A \wedge x \in B))$ (Definition of χ_A, χ_B)
$\Leftrightarrow \forall x \in \mathbf{U}, ((x \notin A \vee x \in A) \wedge (x \notin A \vee x \in B))$ (E14)
$\Leftrightarrow \forall x \in \mathbf{U}, (true \wedge (x \notin A \vee x \in B))$ (E10, E2)
$\Leftrightarrow \forall x \in \mathbf{U}, (x \notin A \vee x \in B)$ (E10, E5)
$\Leftrightarrow \forall x \in \mathbf{U}, (x \in A \Rightarrow x \in B)$ (E18)
$\Leftrightarrow A \subseteq B$ (Definition of $\subseteq$)

(iv) $\forall x \in \mathbf{U}, \chi_A(x) = \chi_B(x) \Leftrightarrow \forall x \in \mathbf{U}, (\chi_A(x) \leq \chi_B(x)) \wedge (\chi_B(x) \leq \chi_A(x))$
$\Leftrightarrow (\forall x \in \mathbf{U})(\chi_A(x) \leq \chi_B(x)) \wedge (\forall x \in \mathbf{U})(\chi_B(x) \leq \chi_A(x))$ (FE9)
$\Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$ (Definition of $\subseteq$)
$\Leftrightarrow A = B$ (Theorem 3.2.1(ii))

(v) $\forall x \in \mathbf{U},$
$\chi_{A \cup B}(x) = 1$
$\Leftrightarrow x \in A \cup B$ (Definition of $\chi_{A \cup B}$)
$\Leftrightarrow x \in A \vee x \in B$ (Definition of $\cup$)
$\Leftrightarrow \chi_A(x) = 1 \vee \chi_B(x) = 1$ (Definition of χ_A, χ_B)
$\Leftrightarrow (\chi_A(x) = 1 \wedge true) \vee (\chi_B(x) = 1 \wedge true)$ (E5)
$\Leftrightarrow (\chi_A(x) = 1 \wedge (\chi_B(x) = 1 \vee \chi_B(x) \neq 1)) \vee (\chi_B(x) = 1 \wedge (\chi_A(x) = 1 \vee \chi_A(x) \neq 1))$ (E2)
$\Leftrightarrow (\chi_A(x) = 1 \wedge (\chi_B(x) = 1 \vee \chi_B(x) = 0)) \vee (\chi_B(x) = 1 \wedge (\chi_A(x) = 1 \vee \chi_A(x) = 0))$
($\chi_S(x) \in \{0,1\}, \forall S \subseteq \mathbf{U}$)
$\Leftrightarrow (\chi_A(x) = 1 \wedge \chi_B(x) = 1) \vee (\chi_A(x) = 1 \wedge \chi_B(x) = 0) \vee$
$(\chi_B(x) = 1 \wedge \chi_A(x) = 1) \vee (\chi_B(x) = 1 \wedge \chi_A(x) = 0)$ (E13)
$\Leftrightarrow (\chi_A(x) = 1 \wedge \chi_B(x) = 1) \vee (\chi_A(x) = 1 \wedge \chi_B(x) = 0) \vee$
$(\chi_A(x) = 1 \wedge \chi_B(x) = 1) \vee (\chi_A(x) = 0 \wedge \chi_B(x) = 1)$ (E9)
$\Leftrightarrow (\chi_A(x) = 1 \wedge \chi_B(x) = 1) \vee (\chi_A(x) = 1 \wedge \chi_B(x) = 0) \vee (\chi_A(x) = 0 \wedge \chi_B(x) = 1)$
(E10, E12, E4)
$$\Leftrightarrow \chi_A(x) + \chi_B(x) - \chi_A(x) \cdot \chi_B(x) = \begin{cases} 1+1-(1\cdot 1) = 1 & \chi_A(x) = 1 \wedge \chi_B(x) = 1; \\ 1+0-(1\cdot 0) = 1 & \chi_A(x) = 1 \wedge \chi_B(x) = 0; \\ 0+1-(0\cdot 1) = 1 & \chi_A(x) = 0 \wedge \chi_B(x) = 1 \end{cases}$$
$\Leftrightarrow \chi_A(x) + \chi_B(x) - \chi_A(x) \cdot \chi_B(x) = 1$
Hence, $\forall x \in \mathbf{U}, \chi_{A \cup B}(x) = \chi_A(x) + \chi_B(x) - \chi_A(x) \cdot \chi_B(x)$.

(vi) $\forall x \in \mathbf{U}, \chi_{A \cap B}(x) = 1 \Leftrightarrow x \in (A \cap B)$ (Definition of $\chi_{A \cap B}$)

$\Leftrightarrow x \in A \wedge x \in B$ (Definition of $\cap$)
$\Leftrightarrow \chi_A(x) = 1 \wedge \chi_B(x) = 1$ (Definition of χ_A, χ_B)
$\Leftrightarrow \chi_A(x) \cdot \chi_B(x) = 1$ $(\forall a, b \in \{0,1\}, a \cdot b = 1 \Leftrightarrow (a = 1) \wedge (b = 1))$

Hence, $\forall x \in \mathbf{U}, \chi_{A \cap B}(x) = \chi_A(x) \cdot \chi_B(x)$. ■

The characteristic functions provide an alternate way for proving set identities.

Example 6.5.24

(i) Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution: $\forall x \in \mathbf{U}, \chi_{A \cap (B \cup C)}(x) = \chi_A(x) \cdot \chi_{(B \cup C)}(x)$ (Lemma 6.5.1(vi))
$= \chi_A(x)(\chi_B(x) + \chi_C(x) - \chi_B(x) \cdot \chi_C(x))$ (Lemma 6.5.1(v))
$= \chi_A(x)\chi_B(x) + \chi_A(x)\chi_C(x) - \chi_A(x)\chi_B(x)\chi_C(x)$
$= \chi_A(x)\chi_B(x) + \chi_A(x)\chi_C(x) - (\chi_A(x)\chi_B(x))(\chi_A(x)\chi_C(x))$
$= \chi_{A \cap B}(x) + \chi_{A \cap C}(x) - \chi_{A \cap B}(x)\chi_{A \cap C}(x)$
$= \chi_{(A \cap B) \cup (A \cap C)}(x)$

By Lemma 6.5.1(iv), $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. □

(ii) Prove that $A \subseteq B$ iff $(A \cap B) = A$.

Solution: $(A \cap B) = A \Leftrightarrow \forall x \in \mathbf{U}, \chi_{A \cap B}(x) = \chi_A(x)$ (Lemma 6.5.1(iv))
$\Leftrightarrow \forall x \in \mathbf{U}, \chi_A(x)\chi_B(x) = \chi_A(x)$ (Lemma 6.5.1(vi))
$\Leftrightarrow \forall x \in \mathbf{U}, \chi_A(x)(1 - \chi_B(x)) = 0$
$\Leftrightarrow \forall x \in \mathbf{U}, (\chi_A(x) = 0 \vee \chi_B(x) = 1)$
$\Leftrightarrow \forall x \in \mathbf{U}, ((\chi_A(x) = 0 \wedge true) \vee (true \wedge \chi_B(x)) = 1)$ (E5, E10)
$\Leftrightarrow \forall x \in \mathbf{U}, ((\chi_A(x) = 0 \wedge (\chi_B(x) \neq 0 \vee \chi_B(x) = 0)) \vee$
$((\chi_A(x) \neq 0 \vee \chi_A(x) = 0) \wedge \chi_B(x) = 1)$ (E2)
$\Leftrightarrow \forall x \in \mathbf{U}, ((\chi_A(x) = 0 \wedge (\chi_B(x) = 1 \vee \chi_B(x) = 0)) \vee$
$((\chi_A(x) = 1 \vee \chi_A(x) = 0) \wedge \chi_B(x) = 1)$ $(\chi_S \in \{0,1\}, \forall S \in \mathbf{U})$
$\Leftrightarrow \forall x \in \mathbf{U}, (((\chi_A(x) = 0 \wedge \chi_B(x) = 1) \vee (\chi_A(x) = 0 \wedge \chi_B(x) = 0)) \vee$
$((\chi_A(x) = 1 \wedge \chi_B(x) = 1) \vee (\chi_A(x) = 0 \wedge \chi_B(x) = 1))$ (E13, E10)
$\Leftrightarrow \forall x \in \mathbf{U}, (((\chi_A(x) = 0 \wedge \chi_B(x) = 1) \vee (\chi_A(x) = 0 \wedge \chi_B(x) = 0)) \vee$
$((\chi_A(x) = 1 \wedge \chi_B(x) = 1))$ (E4)
$\Leftrightarrow \forall x \in \mathbf{U}, \chi_A(x) \leq \chi_B(x)$
$\Leftrightarrow A \subseteq B$ (Lemma 6.5.1(iii)) ■

6.6 Exercises

1. Let $\mathbf{R}$ be the set of all real numbers and $f : \mathbf{R} \rightarrow \mathbf{R}$ such that $f(x) = |x|, \forall x \in \mathbf{R}$. Is f one-to-one? Is f onto?
2. Let $\mathbf{N}$ be the set of positive integers. Let $g : \mathbf{N} \times \mathbf{N} \rightarrow \mathbf{N}$ such that
$$g((i,j)) = 2^i 3^j$$
Prove that g is a one-to-one function. Is it onto?
3. Let $f : \mathbf{R} - \{-\frac{1}{2}\} \rightarrow \mathbf{R} - \{\frac{1}{2}\}$ such that
$$f(x) = \frac{x-3}{2x+1}, \forall x \in \mathbf{R} - \{-\tfrac{1}{2}\}$$
Prove that f is one-to-one and onto.

4. Let $f : X \to Y$ and $g : X \to Y$. Prove that if $f \subseteq g$, then $f = g$
5. Let $f : X \to X$. Prove that:
   (a) if $f \subseteq I_X$, then $f = I_X$.
   (b) if $I_X \subseteq f$, then $f = I_X$.
6. Prove that if $A \subseteq B$, then $A^C \subseteq B^C$ for any set C.
7. Prove that $X^Y = Y^X$ iff $X = Y$.
8. Prove that $Y^X = \emptyset$ iff $Y = \emptyset$ and $X \neq \emptyset$.
9. Let f be defined as in Example 6.1.18(vii). Let $A = \{(a, b) \in \mathbf{R} \times \mathbf{R} \mid -1 \leq a, b \leq 1\}$. Determine $f(A)$.
10. Let $f : X \to Y$. Prove that f is onto iff $\forall B \subseteq Y, B \neq \emptyset \Rightarrow f^{-1}(B) \neq \emptyset$.
11. Let $f : X \to Y$ be such that $f(X) = Y$. Let $B \subseteq Y$ and $C \subseteq Y$. Prove that if $f^{-1}(B) = f^{-1}(C)$, then $B = C$.

    Give an example to show that the assertion is false if $f(X) \neq Y$.
12. Let $f : X \to Y$. Prove that $f(A \cap B) = f(A) \cap f(B)$, $\forall A, B \subseteq X$ iff f is one-to-one.
13. Let $f : X \to Y$ be a one-to-one correspondence. Prove that:
    (a) $f \circ f^{-1} = I_X$, where I_X is the identity functions in X.
    (b) $f^{-1} \circ f = I_Y$, where I_Y is the identity functions in Y.
14. Let $f : X \to Y$. If there exist $g : Y \to X$ and $h : Y \to X$ such that $f \circ g = I_X$ and $h \circ f = I_Y$, Prove that:
    (a) the function f is a one-to-one correspondence.
    (b) $g = h = f^{-1}$.
15. Let $X \neq \emptyset$. Define $\mathcal{R} = \{(f, g) \in X^X \times X^X \mid Ran(f) = Ran(g)\}$. Prove that $\mathcal{R}$ is an equivalence relation in X^X.
16. Let $\mathbf{R}$ be the set of real numbers and $\mathcal{F}$ be the set of all functions from $\mathbf{R}$ to $\mathbf{R}$. Define:
$$\lesssim = \{(f, g) \in \mathcal{F} \times \mathcal{F} \mid f(x) \leq g(x), \forall x \in \mathbf{R}\}.$$
    Prove that $(\mathcal{F}, \lesssim)$ is a partially ordered set.
17. Let $(X, \lesssim)$ and $(Y, \preceq)$ be a partially ordered sets and $f : X \overset{onto}{\to} Y$. Let $x \lesssim y$ iff $f(x) \preceq f(y)$, $\forall x, y \in X$. Prove that f is one-to-one.
18. Prove each of the following by using charateristic functions.
    (a) $\forall x, x \in A \Leftrightarrow x \notin \overline{A}$
    (b) $\overline{\overline{A}} = A$
    (c) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
19. Let $f : X \to Y$.
    (a) Let R be a relation in X such that $(x, y) \in R$ iff $f(x) = f(y)$. Prove that R is an equivalence relation.
    (b) Let $j : X \to [X]/R$ such that $j(x) = [x]/R, \forall x \in X$. Prove that j is onto.
    (c) Let g be a relation from $[X]/R$ to $Ran(f)$ such that $(x, y) \in g \Leftrightarrow y = f(w) \wedge x = [w]/R, \forall x \in [X]/R$. Prove that:
        - g is a function from $[X]/R$ to $Ran(f)$;
        - g is one-to-one and onto.
    (d) Let $i : Ran(f) \to Y$ such that $i(y) = y, \forall y \in Ran(f)$. Prove that i is one-to-one.
    (e) Prove that $f = j \circ g \circ i$.